THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

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CLASS 10 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 2

Chapter 14: **Probability** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. A bag contains 4 red balls and x blue balls. The probability of drawing a blue ball is 3 times the probability of drawing a red ball. The value of x is: 1

(a) 16 **(b)** 12 **(c)** 11 **(d)** 3

2. A bag contains 4 red and 6 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{5}{8}$. The value of x is: 1

(a) 6 **(b)** 12 **(c)** 7 **(d)** 5

3. A red die and a blue die are rolled together. The probability that at least one die shows 6 is: 1

(a) $\frac{1}{3}$ **(b)** $\frac{11}{36}$ **(c)** $\frac{25}{36}$ **(d)** $\frac{5}{18}$

4. A letter is chosen at random from the word 'PROBABILITY'. The probability that it is a vowel is: 1

(a) $\frac{7}{11}$ **(b)** $\frac{4}{11}$ **(c)** $\frac{3}{11}$ **(d)** $\frac{5}{11}$

5. A red die and a blue die are rolled together. The probability that the sum is 12 is: 1

(a) $\frac{1}{18}$ **(b)** 0 **(c)** $\frac{1}{36}$ **(d)** $\frac{35}{36}$

6. A coin is tossed three times. The probability of getting heads on the first and last tosses is: 1

(a) $\frac{1}{8}$ **(b)** $\frac{3}{8}$ **(c)** $\frac{1}{4}$ **(d)** $\frac{3}{4}$

7. 50 cards numbered 1 to 50 are placed in a box, mixed well and one is drawn at random. The probability that the number drawn is a multiple of 2 but not of 4 is: 1

(a) $\frac{7}{25}$ **(b)** $\frac{13}{50}$ **(c)** $\frac{6}{25}$ **(d)** $\frac{37}{50}$

8. A bag contains 5 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. The value of x is: 1

(a) 15 **(b)** 9 **(c)** 10 **(d)** 2

9. A bag contains 5 red and 9 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{4}{7}$. The value of x is: 1

(a) 8 **(b)** 7 **(c)** 14 **(d)** 6

10. A red die and a blue die are rolled together. The probability that the sum is 3 or 11 is: 1

(a) $\frac{8}{9}$ **(b)** $\frac{1}{9}$ **(c)** $\frac{5}{36}$ **(d)** $\frac{1}{12}$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. 42 cards numbered 1 to 42 are placed in a box, mixed well and one is drawn at random. Find the probability that the number drawn is a prime number. 2

12. A bag contains 5 red, 9 blue, 6 green balls. One ball is drawn at random. Find the probability that it is red or blue. 2

13. A red die and a blue die are rolled together. Find the probability that the product is less than 8. 2

14. 48 cards numbered 1 to 48 are placed in a box, mixed well and one is drawn at random. Find the probability that the number drawn is a perfect square. 2

SECTION — C (Short Answer Type) — 4 × 3 = 12

Answer in about 40–60 words each.

15. A bag contains 9 red balls and x blue balls. The probability of drawing a blue ball is 2 times the probability of drawing a red ball. Find the value of x. 3

16. A bag contains 4 red and 12 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{2}{5}$. Find the value of x. 3

17. A red die and a blue die are rolled together. Find the probability that both dice show the same number (a doublet). 3

OR

A bag contains 10 red balls and x blue balls. The probability of drawing a blue ball is 4 times the probability of drawing a red ball. Find the value of x. 3

18. A bag contains 7 red and 3 blue balls. When x more red balls are added, the probability of drawing a red ball becomes $\frac{17}{20}$. Find the value of x. 3

OR

A red die and a blue die are rolled together. Find the probability that the product of the numbers is 12. 3

SECTION — D (Long Answer Type) — 2 × 5 = 10

Answer in about 100–150 words each.

19. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) the sum of the numbers is 8,
(ii) the first number is greater than the second,
(iii) at least one die shows 6 (in this order).

OR

A box contains 36 cards numbered 1 to 36. One card is drawn at random. Find the probability that the number is: 5
(i) divisible by both 2 and 3,
(ii) a multiple of 3 or 5,
(iii) a composite number (in this order).

20. Two dice are thrown together. Find the probabilities of these events (in this order): 5
(i) both dice show the same number (a doublet),
(ii) the sum is 3 or 11,
(iii) neither die shows a 3 (in this order).

OR

A box contains 60 cards numbered 1 to 60. One card is drawn at random. Find the probability that the number is: 5
(i) a multiple of 4,
(ii) not a multiple of 7,
(iii) greater than 12 (in this order).

END OF QUESTION PAPER — Total: 40 Marks